Fitting logistic regression models

Logistics and reminders

- **Optional** HW 6 due on Friday (4/24)
- Project Part 4 due Wednesday 4/29 (LDOC)
- Project work time in class after we finish up logistic regression

Last time: Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

What's the probability of the observed data, if $\beta_0 = 3$ and $\beta_1 = -0.5$?

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

Length	3	4	5	6	7
π	$\frac{\exp\{3 - 0.5(3)\}}{1 + \exp\{3 - 0.5(3)\}} = 0.818$	0.731	0.622	0.5	0.378

$L(\beta_0, \beta_1) = P(\text{data}) =$

$$(0.818)^3(0.731)^2(0.622)^3(1 - 0.622)^2(0.5)(1 - 0.5) \\ \cdot (0.378)(1 - 0.378)^2$$

$= 0.000368$

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

What's the probability of the observed data, if $\beta_0 = 4$ and $\beta_1 = -0.75$?

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

Length	3	4	5	6	7
π	$\dfrac{\exp\{4 - 0.75(3)\}}{1 + \exp\{4 - 0.75(3)\}} = 0.852$	0.731	0.562	0.378	0.223

$$L(\beta_0, \beta_1) = P(\text{data}) =$$

$$(0.852)^3(0.731)^2(0.562)^3(1 - 0.562)^2(0.378)(1 - 0.378) \cdot (0.223)(1 - 0.223)^2$$

$$= 0.000356$$

Maximum likelihood for logistic regression

Likelihood:

- For estimates $\widehat{\beta}_0$ and $\widehat{\beta}_1$, $\widehat{\pi} = \dfrac{\exp\{\widehat{\beta}_0 + \widehat{\beta}_1 x\}}{1 + \exp\{\widehat{\beta}_0 + \widehat{\beta}_1 x\}}$
- $L(\widehat{\beta}_0, \widehat{\beta}_1) = P(\text{data})$

Maximize:

- Choose $\widehat{\beta}_0, \widehat{\beta}_1$ to maximize $L(\widehat{\beta}_0, \widehat{\beta}_1)$

> In the class activity examples, we have only considered a few different possibilities for $\widehat{\beta}_0$ and $\widehat{\beta}_1$. In reality, we consider all real numbers as possible values.

Fitting logistic regression

Logistic regression with maximum likelihood

- **Model:** $\log\left(\dfrac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 x$
- **Measure of fit:** $L(\widehat{\beta}_0, \widehat{\beta}_1) =$ probability of observed data for estimates $\widehat{\beta}_0, \widehat{\beta}_1$
- **Optimize:** choose $\widehat{\beta}_0, \widehat{\beta}_1$ to maximize $L(\widehat{\beta}_0, \widehat{\beta}_1)$

Linear regression with least squares

- **Model:** $y = \beta_0 + \beta_1 x + \varepsilon$
- **Measure of fit:** $SSE = \sum_{i=1}^{n} (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i)^2$
- **Optimize:** choose $\widehat{\beta}_0, \widehat{\beta}_1$ to minimize SSE

Regression assumptions

What assumptions did we make for linear regression models?

Regression assumptions

What assumptions did we make for linear regression models?

- Shape
- Constant variance
- Normality
- Independence
- Randomness

Not all these assumptions translate to logistic regression.

Assumptions: logistic vs. linear regression

Logistic regression:

$$\pi = P(y = 1)$$

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 x$$

- shape
- randomness
- independence

Linear regression:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

$$\varepsilon \sim N(0, \sigma_\varepsilon^2)$$

- shape
- randomness
- independence
- constant variance
- normality

> The constant variance and normality assumptions apply specifically to the noise term ε

Data

Relationship between the length of a putt, and whether it was made:

- $x =$ length (feet)
- $y =$ result (success = 1, failure = 0)

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 x$$

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134

Assessing the shape assumption

> How did we assess the shape assumption for linear regression?

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With scatterplots and residual plots.

> Can we use these diagnostics for logistic regression?

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With scatterplots and residual plots.

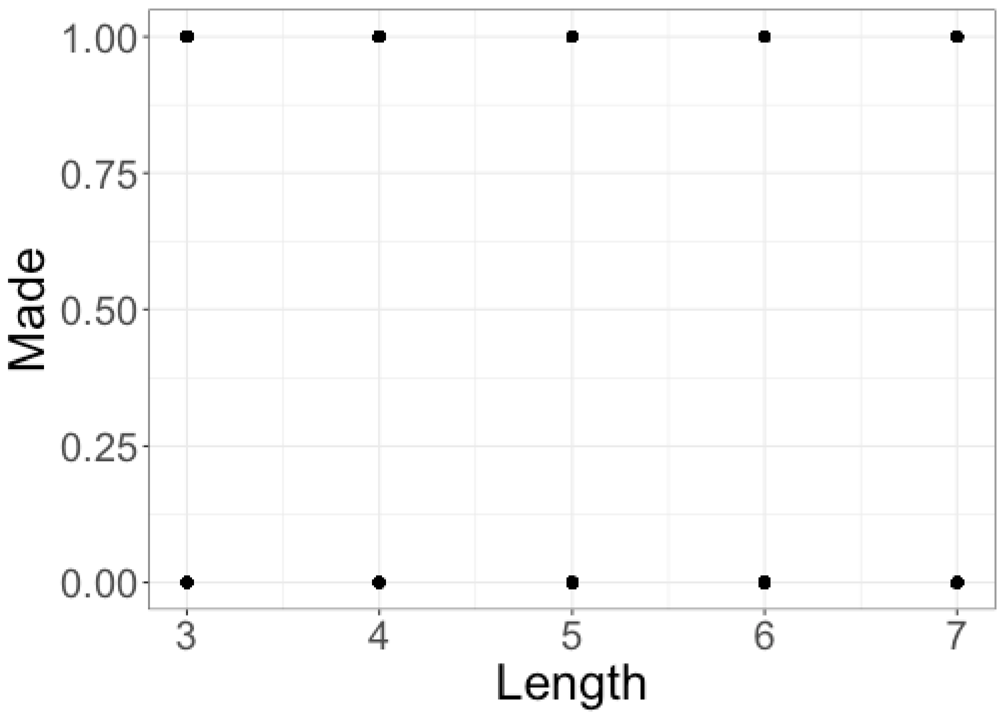
> Can we use these diagnostics for logistic regression?

No:

- the response is binary
- the shape assumption is that the *log odds* (not the response) are a linear function of the predictor(s)

Assessing the shape assumption

Scatterplots don't work for logistic regression:



Shape

- Shape assumption is that the log-odds are (at least approximately) a linear function of x: $\log(odds) \approx \beta_0 + \beta_1 x$
- We can assess the shape assumption by plotting the **empirical log-odds** (aka empirical logits)
- Example: relationship between the length of a putt and whether it was made

Length	**3**	**4**	**5**	**6**	**7**
Number of successes	84	88	61	61	44
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Empirical log-odds

Step 1: estimate the probability of success for each length of putt

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134
Probability of success $\hat{p}$	0.832	0.739	0.565	0.488	0.328

Empirical log-odds

Step 2: convert empirical probabilities to empirical odds

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134
Probability of success $\hat{p}$	0.832	0.739	0.565	0.488	0.328
Odds $\dfrac{\hat{p}}{1-\hat{p}}$	4.941	2.839	1.298	0.953	0.489

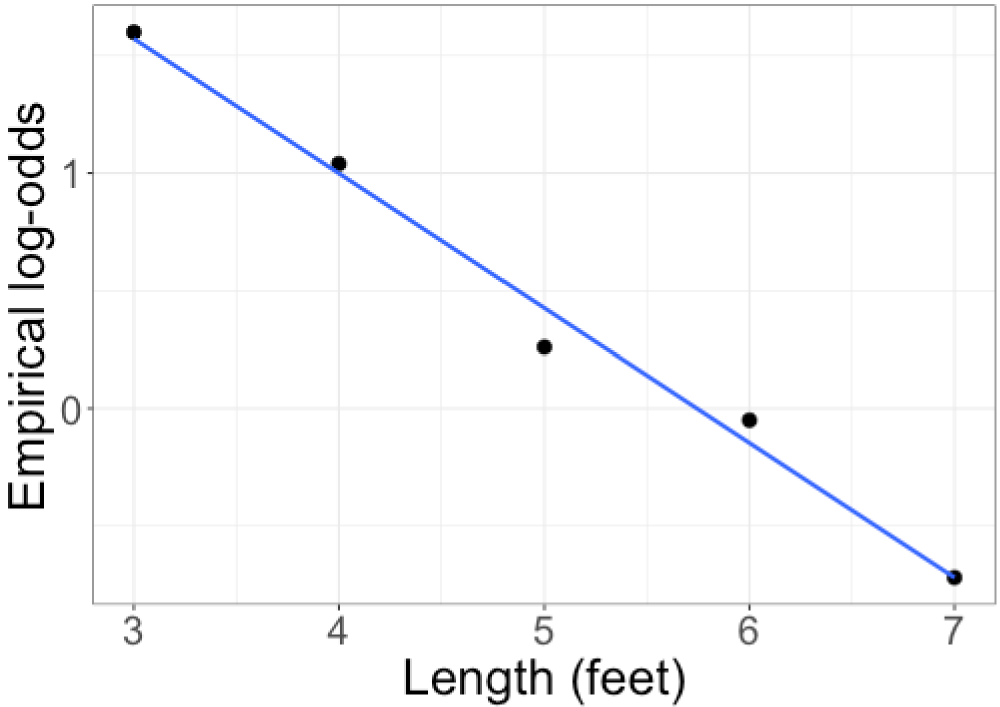
Empirical log-odds

Step 3: calculate empirical log-odds

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134
Probability of success $\hat{p}$	0.832	0.739	0.565	0.488	0.328
Odds $\dfrac{\hat{p}}{1-\hat{p}}$	4.941	2.839	1.298	0.953	0.489
Log-odds $\log\left(\dfrac{\hat{p}}{1-\hat{p}}\right)$	1.60	1.04	0.26	-0.05	-0.72

Empirical log-odds

Step 4: plot empirical log-odds against predictor



> The shape assumption is reasonable if the empirical log-odds plot looks roughly linear.

Empirical log odds plots

Now let's try this with the med school admissions data:

GPA	2.26	2.42	2.48	2.52	2.55	2.56	...
Admit = 0	1	1	1	1	1	1	...
Admit = 1	0	1	0	0	0	0	...

> What problem do I run into?

Empirical log odds plots

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Admit = 1	0	1	0	0	0	0	...

> What problem do I run into?

There are a small number of observations for each GPA, so we can't calculate empirical log odds for each GPA.

Solution: Divide the data into *bins*

Empirical log odds plots

- Divide observations into bins of equal size, based on values of predictor X
- Calculate empirical log odds in each bin
- Plot empirical log odds